

DNA Replication in Prokaryotes and Eukaryotes

A Complete Molecular Biology Teaching and Learning Module

Cheat Sheet

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DNA Replication — Quick Reference Cheat Sheet

Course: Molecular Biology of DNA Replication

Essential Terminology

Term	Definition
Semiconservative	Each daughter molecule = 1 old strand + 1 new strand
Origin (ori)	DNA sequence where replication initiates
Replication fork	Y-shaped structure where DNA is unwound and synthesized
Leading strand	Synthesized continuously toward the fork (5'→3')
Lagging strand	Synthesized discontinuously away from the fork (5'→3')
Okazaki fragments	Short DNA fragments on the lagging strand
Processivity	Nucleotides added before polymerase dissociates
Primer	Short RNA/DNA that provides free 3'-OH for polymerase
Replisome	Complete replication machinery at the fork
Telomere	Protective structure at chromosome ends
Licensing	Assembly of pre-RC to allow origin firing

Replication Proteins — Quick Function Table

E. coli (Prokaryotic)

Protein	Gene	Function
DnaA	dnaA	Initiator; binds oriC
DnaB	dnaB	Helicase (unwinds DNA)
DnaC	dnaC	Helicase loader
DnaG	dnaG	Primase (makes RNA primer)
Pol III	dnaE/Q/X	Main replicative polymerase
ε subunit	dnaQ	Proofreading (3'→5' exonuclease)
β clamp	dnaN	Processivity factor (dimer)
γ complex	dnaX	Clamp loader
SSB	ssb	Single-strand binding protein

Protein	Gene	Function
Gyrase	gyrA/B	Type II topoisomerase (introduces negative supercoils)
Topo I	topA	Type I topoisomerase
Topo IV	parC/E	Decatenation
Pol I	polA	Primer removal + gap filling
Ligase	ligA	Seals nicks (NAD ⁺ -dependent)
MutS/MutL/MutH	mutS/L/H	Mismatch repair

Human (Eukaryotic)

Protein	Function
ORC (Orc1-6)	Origin recognition complex
Cdc6	Recruits MCM; ATPase
Cdt1	Loads MCM2-7
MCM2-7	Replicative helicase (inactive until activated)
Cdc45	CMG helicase component
GIN5	CMG helicase component
RPA	Single-strand binding protein
Pol α/primase	Initiates synthesis; makes RNA-DNA primer
Pol ϵ	Leading strand synthesis
Pol δ	Lagging strand synthesis + repair
PCNA	Sliding clamp (homotrimer)
RFC	Clamp loader
RNase H2	Removes RNA primers
FEN1	Flap endonuclease; primer processing
DNA Ligase I	Seals nicks (ATP-dependent)
Topo I	Relaxes positive supercoils
Topo IIα	Relieves supercoiling + decatenation
MSH2-6/MLH1-PMS2	Mismatch repair
ATR/CHK1	Replication checkpoint
TERT/TERC	Telomerase (telomere maintenance)
Shelterin	Telomere protection complex
Geminin	Inhibits Cdt1 (prevents re-replication)

Replication Sequence — Prokaryotic (E. coli)

1. DnaA-ATP binds oriC → AT-rich region melts
2. DnaC loads DnaB helicase (2 hexamers)
3. DnaG primase synthesizes RNA primers
4. Clamp loader loads β sliding clamp
5. Pol III synthesizes DNA:
 - Leading strand: continuous, one primer
 - Lagging strand: Okazaki fragments (1-2 kb), multiple primers
6. Proofreading: ϵ subunit (3'→5' exonuclease)
7. Pol I: removes RNA primers, fills gaps
8. DNA ligase: seals nicks
9. Gyrase: relieves supercoiling ahead of fork
10. Tus/Ter: termination at Ter region
11. Topo IV: decatenates daughter chromosomes

Replication Sequence — Eukaryotic (Human)

- G1 PHASE (LICENSING):
1. ORC binds origins
 2. Cdc6 + Cdt1 recruited
 3. MCM2-7 double hexamers loaded → Origin licensed
- S PHASE (ACTIVATION):
4. CDK + DDK phosphorylate MCM
 5. Cdc45 + GINS loaded → CMG helicase active
 6. CMG unwinds DNA
 7. RPA binds ssDNA
 8. Pol α /primase makes RNA-DNA primer
 9. RFC loads PCNA
 10. Polymerase switching: Pol α → Pol ϵ (leading) or Pol δ (lagging)
 11. Leading: Pol ϵ continuous; Lagging: Pol δ , Okazaki fragments (100-200 nt)
- PROCESSING:
12. RNase H2 removes RNA primers
 13. FEN1 removes flaps
 14. Pol δ fills gaps
 15. DNA Ligase I seals nicks
 16. CAF-1 reassembles nucleosomes
- TERMINATION:
17. Forks converge → CMG disassembled (p97/Cdc48)
 18. Gaps filled → ligation
- TELOMERES:
19. Telomerase extends 3' overhang (TERT + TERC)
 20. Complementary strand filled by Pol α /Pol δ

Leading vs. Lagging Strand

Feature	Leading Strand	Lagging Strand
Direction	5'→3' toward fork	5'→3' away from fork
Synthesis	Continuous	Discontinuous

Feature	Leading Strand	Lagging Strand
Primers	One	Multiple (per Okazaki fragment)
Okazaki fragments	No	Yes
Speed	Equal to fork speed	Equal to fork speed (overall)
E. coli polymerase	Pol III	Pol III (with Pol I for processing)
Human polymerase	Pol ϵ	Pol δ (with Pol α for priming)

Okazaki Fragment Processing

Step	E. coli	Eukaryotes
Fragment size	1,000–2,000 nt	100–200 nt
Primer removal	Pol I (5'→3' exo)	RNase H2 + FEN1
Gap filling	Pol I	Pol δ
Ligation	Ligase (NAD ⁺)	Ligase I (ATP)

Major Differences: Prokaryotes vs. Eukaryotes

Feature	Prokaryotes	Eukaryotes
Genome	Circular, 4.6 Mb	Linear, 3.2 Gb
Origins	1 (oriC)	30,000–50,000
Helicase	DnaB (3'→5')	CMG (5'→3')
Sliding clamp	β_2 (dimer)	PCNA (trimer)
Leading Pol	Pol III	Pol ϵ
Lagging Pol	Pol III	Pol δ
Primase	DnaG	Pol α /primase
Okazaki size	1–2 kb	100–200 bp
Fork speed	1,000 nt/s	50 nt/s
Telomeres	No	Yes
Chromatin	No (nucleoid)	Yes (nucleosomes)
Licensing	No	Pre-RC (ORC/Cdc6/Cdt1/MCM)
MMR strand discrimination	Dam methylation	Strand nicks/PCNA
Termination	Tus/Ter	Fork convergence

Important Polymerases

Polymerase	Direction	Proofreading	Function
E. coli Pol I	5'→3'	Yes (3'→5')	Primer removal, repair
E. coli Pol III	5'→3'	Yes (ε subunit)	Main replication
Human Pol α	5'→3'	No	Primer synthesis
Human Pol δ	5'→3'	Yes	Lagging strand, repair
Human Pol ε	5'→3'	Yes	Leading strand
Human Pol γ	5'→3'	Yes	Mitochondrial DNA

Telomere Quick Facts

- **Sequence:** TTAGGG (human), repeated 2,000-3,000 times
- **Length:** 5-15 kb at birth; shortens ~50-200 bp/division
- **Enzyme:** Telomerase = TERT (protein) + TERC (RNA template)
- **Template in TERC:** 3'-CAAUCCCAAUC-5'
- **Shelterin:** TRF1, TRF2, POT1, TIN2, TPP1, RAP1
- **Cancer:** 85-90% reactivate telomerase; 10-15% use ALT
- **Aging:** Short telomeres → senescence (Hayflick limit ~50-60 divisions)

Checkpoints

Checkpoint	Sensor	Effector	Trigger
ATR-CHK1	ATRIP	CHK1	ssDNA at stalled forks
ATM-CHK2	MRN	CHK2	DSBs
Intra-S	ATR	CHK1	Replication stress
G2/M	ATM/ATR	CHK1/CHK2	Unreplicated DNA, damage
Spindle	Mad2	—	Unattached kinetochores

Important Exam Facts

1. DNA is **always** synthesized 5'→3' (NO exceptions)
2. **Both** strands are synthesized 5'→3' (lagging strand = Okazaki fragments)
3. DNA polymerase **cannot** start synthesis without a primer
4. E. coli ligase uses **NAD⁺**; eukaryotic ligase uses **ATP**
5. The sliding clamp increases processivity from ~20 to >50,000 nt
6. E. coli has **one origin** (oriC); humans have **~30,000-50,000**
7. Human Okazaki fragments: **100-200 nt**; E. coli: **1-2 kb**
8. Telomerase is a **reverse transcriptase** (uses RNA template)

9. **Fluoroquinolones** target DNA gyrase (bacteria-specific)
10. **Lynch syndrome** = defective mismatch repair → MSI
11. Proofreading improves fidelity **100-fold** ($10^{-5} \rightarrow 10^{-7}$)
12. MMR improves fidelity another **100-1000-fold** ($10^{-7} \rightarrow 10^{-9}-10^{-10}$)
13. **Licensing** (G₁) and **firing** (S) are mutually exclusive (CDK)
14. **PCNA** is a **trimer**; **β clamp** is a **dimer**
15. Fork speed: E. coli ~1,000 nt/s; Human ~50 nt/s

Common Mistakes to Avoid

Mistake	Correction
"Lagging strand is synthesized 3'→5'"	Both strands are 5'→3'
"DNA polymerase starts from scratch"	Needs a primer (3'-OH)
"Okazaki fragments are the same size in all organisms"	E. coli: 1-2 kb; Human: 100-200 bp
"Proofreading is the main fidelity mechanism"	All three layers contribute equally important
"Telomerase is active in all cells"	Inactive in most somatic cells
"Both strands need the same number of primers"	Leading: 1 primer; Lagging: many
"Replication and transcription use the same polymerase"	Completely different polymerases
"PCR is the same as cellular replication"	PCR uses heat denaturation, not helicase